

Precalculus Review for Calculus



Functions and composition

- 1 Let $f(x) = \sqrt{x-1}$ and $g(x) = x^2 + 3$. Find:
 - a $(f \circ g)(x)$
 - b $(g \circ f)(x)$
 - c the domain of $f \circ g$
 - 2 Simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$ for $f(x) = x^2 - 3x$.
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Lines, exponentials, and trig — the toolkit calculus builds on

- 3 Find the slope and equation of the line through $(-1, 4)$ and $(3, -2)$.
 - 4 Solve $2e^{3x} = 14$ exactly and to three decimal places.
 - 5 State the exact value of each expression, needed frequently when differentiating trig functions:
 - a $\sin \frac{\pi}{4}$
 - b $\cos \frac{\pi}{3}$
 - c $\tan \frac{\pi}{6}$
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Factoring and rational expressions

- 6 Factor each expression fully:
 - a $x^2 - 5x + 6$
 - b $4x^2 - 9$
 - c $x^3 - 8$
 - 7 Simplify $\frac{x^2 - 4}{x^2 - x - 6}$, stating the value(s) of x excluded from the domain.
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Inverse functions and logarithms

- 8 Find the inverse of $f(x) = 3x - 5$, and verify by computing $f(f^{-1}(x))$.
 - 9 Solve $\ln(x - 1) = 2$ exactly and to three decimal places.
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Piecewise and absolute value review

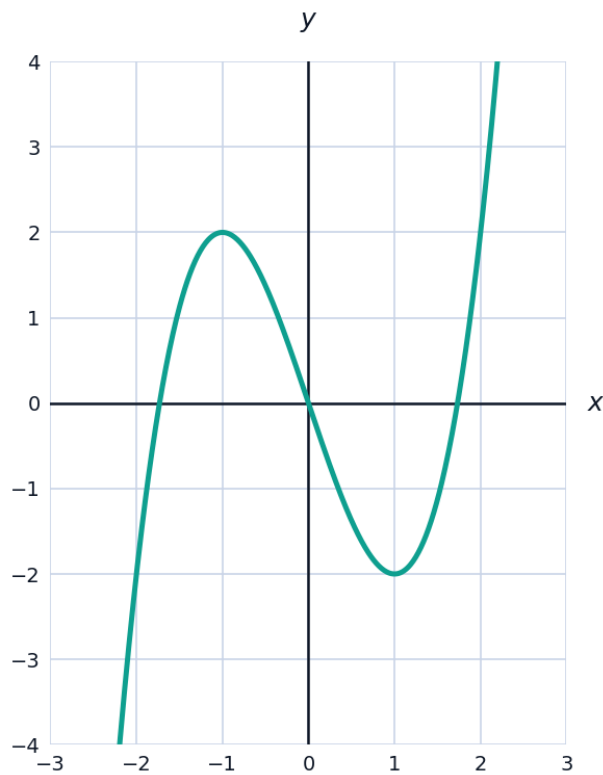
- 10 Sketch-reasoning: rewrite $f(x) = |2x - 6|$ as a piecewise function without absolute value bars.
- 11 Solve the inequality $|3x + 1| \leq 7$, writing your answer in interval notation.
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Systems of equations

- 12 Solve the system $2x + y = 7$ and $x - y = 2$ by elimination.
- 13 Find the point(s) of intersection of $y = x^2$ and $y = 2x + 3$.
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Graphing functions calculus will build on

- 14 The graph of $f(x) = x^3 - 3x$ is shown. Use it to estimate the x-intercepts and the local max/min locations — features calculus will let you find exactly.



Average rate of change (a preview of the derivative)

- 15 Find the average rate of change of $f(x) = x^2$ over $[1, 4]$, and interpret this as the slope of a secant line.

- 16 For $f(x) = x^2$, find the average rate of change over $[1, 1 + h]$ in terms of h , and describe what happens as $h \rightarrow 0$ (a preview of the derivative).
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Radicals and rational exponents

- 17 Simplify each expression, writing your answer with positive exponents:

a $x^{1/2} \cdot x^{1/3}$

b $(x^{2/3})^3$

c $\frac{x^{5/2}}{x^{1/2}}$

- 18 Rewrite $\sqrt[3]{x^2}$ using rational exponent notation, and evaluate it at $x = 8$.
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Interval notation and set operations review

- 19 Write the solution set of $x^2 - 4 > 0$ in interval notation.